

# The Rosen Fibration

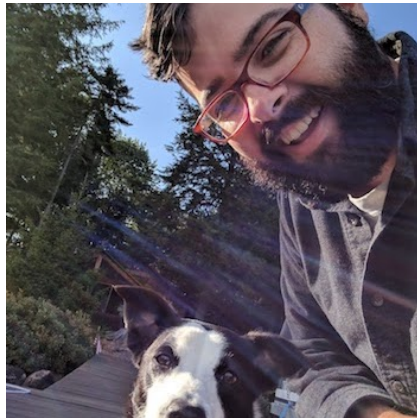
## ACT26 — Tallinn

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# How it started

At the end of October 2025, Amahury (L) and Sean (R, with doggo) contacted me.



- Applying category theory to biology: 'vindicating' Robert Rosen, who believed category theory was a foundation for life sciences.

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- His work foreshadows ideas in endofunctor algebra theory, monads, systems and processes—concepts we understand **only now**.
- This project (with Amahury and Sean) is a work in progress; comments welcome.

# Robert Rosen (1934–1998)

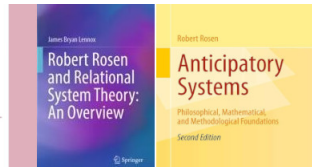


- Attended lectures of Eilenberg, then Mac Lane
- Wrote extensively on category theory in biology
- not always correct...

Wrote **before** Yoneda lemma, adjunctions, monads ... and stayed outside the category theory community.

ESSAYS  
ON  
LIFE  
ITSELF

ROBERT ROSEN



# MR systems

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A **candle** has  $f$  but not  $\Phi$ ; **Ni-Ti alloy** has  $\Phi$  (seals microcracks) but no metabolism.

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Our goal: study the diagram  $A \rightarrow B \rightarrow [A, B]$  in a suitable sense. Can we build a category

$$\sum_{A, B: \mathcal{C}} \{A \xrightarrow{f} B \xrightarrow{\Phi} [A, B]\}?$$



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$$\sum_{A, B: C} \{A \xrightarrow{f} B \xrightarrow{\Phi} [A, B]\}?$$

Yes — **MRS** is the **Rosen fibration** of the title.

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## Working Assumption

$\mathcal{C}$  is symmetric monoidal closed, with all needed (co)limits.

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## Definition

An **incoherent MR system** is  $A \xrightarrow{f} B \xrightarrow{\Phi} [A, B]$  in  $\mathcal{C}$ . A morphism  $(\ell, r) : (f, \Phi) \rightarrow (g, \Psi)$  makes

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{\Phi} & [A, B] \\ \downarrow l & & \downarrow r & & \searrow [A, r] \\ A' & \xrightarrow{g} & B' & \xrightarrow{\Psi} & [A', B'] \\ & & & & \nearrow [\ell, B'] \\ & & & & [A, B'] \end{array}$$

commute. The total category **iMRS** has triples  $(A, B; f, \Phi)$  as objects and morphisms as above.

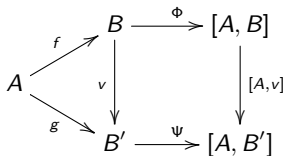
# I use agda, btw

```
1 record iMR2 (A B : Obj) : Set (o ∪ ℓ) where
2   constructor ⟨⟦_,_⟧⟩
3   field
4     f : A ⇒ B
5     φ : B ⇒ [ A , B ]o
6
7 record iMR2o : Set (o ∪ ℓ) where
8   field
9     A B : Obj
10    ξ : iMR2 A B
11
12 record iMR2⇒ (X Y : iMR2o) : Set (o ∪ ℓ ∪ e) where
13   module X = iMR2o X
14   module Y = iMR2o Y
15   module ξX = iMR2 X.ξ
16   module ξY = iMR2 Y.ξ
17   field
18     l : X.A ⇒ Y.A
19     r : X.B ⇒ Y.B
20     eqf : r ∘ ξX.f ≈ ξY.f ∘ l
21     eqφ : [ l , id ]1 ∘ ξY.φ ∘ r ≈ [ id , r ]1 ∘ ξX.φ
```

# Reasoning fiberwise

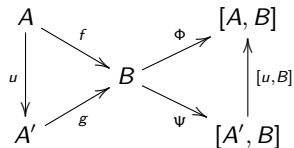
() Fix  $A$ : category  $\mathbf{iMRS}^A$  ( $\ell = \text{id}$ )

- same  $A$ , varying  $B$ ,
- morphisms  $v : B \rightarrow B'$  with



() Fix  $B$ : category  $\mathbf{iMRS}_B$  ( $r = \text{id}$ )

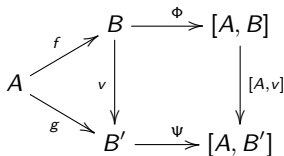
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# Reasoning fiberwise

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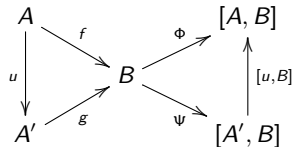
```

1 record iMR2n (A : Obj) : Set (o u l u e) where
2   field
3     B : Obj
4     ξ : iMR2 A B
5
6 record iMR2n → {A : Obj} (X Y : iMR2n A) : Set (o u l u e) where
7   module X = iMR2n X
8   module Y = iMR2n Y
9   module ξX = iMR2 X.ξ
10  module ξY = iMR2 Y.ξ
11  field
12    v : X.B → Y.B
13    eqf : ξX.f = ξY.f ∘ v

```

(☞) Fix  $B$ : category  $\mathbf{iMRS}_B$  ( $r = \text{id}$ )

- same  $B$ , varying  $A$ ,
- morphisms  $u : A \rightarrow A'$  with



```

1 record iMR2l (B : Obj) : Set (o u l u e) where
2   field
3     A : Obj
4     ξ : iMR2 A B
5
6 record iMR2l → {B : Obj} (X Y : iMR2l B) : Set (o u l u e) where
7   module X = iMR2l X
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9   module ξX = iMR2 X.ξ
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12    u : X.A → Y.A
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```

# Incoherent systems

## Equivalent characterization of an incoherent MR system:

- Fix  $A$ ; one can think of the category of iMRS 'based at  $A$ ' as the category of algebras for the endofunctor of  $\mathcal{C}$

$$X \mapsto A \otimes (I + X)$$

- Fix  $B$ ; the category of iMRS 'based at  $B$ ' is the slice  $\mathcal{C}/[I + B, B]$ .

A slightly more refined characterization, that works only in a distributive Cartesian base, is the following:

- An iMR based at  $A$  is a parametric, pointed dynamical system of type  $\xi \approx (1 \xrightarrow{f} B; B \xrightarrow{\phi} B)$  with certain coKleisli maps;
- More simply, an iMR seems to be a **lens**  $(\begin{smallmatrix} A \\ B \end{smallmatrix}) \rightarrow (\begin{smallmatrix} B \\ B \end{smallmatrix})$ . (There is more to this story I haven't looked into...)

# There is a problem!

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$(\text{reindexing}) B \mapsto \mathbf{iMRS}_B$  is **not** functorial; the categories  $\mathbf{iMRS}_B$  are displayed, not fibered, over  $\mathcal{C}$ .

```
1 left : {u : A → A'} → Functor (IMR2A A') (IMR2A A)
2 left {A} {A'} u = record
3   { Fu = λ { x } →
4     let module x = IMR2A x
5     in record
6       { B = x.B
7       ; ξ = λ IMR2.f x.ξ → u ∘ [ u , id ] ∘ IMR2.φ x.ξ }
8   }
9 ; Fv = λ { {X} {Y} } f →
10   let module x = IMR2A x
11   module ξx = IMR2 x.ξ
12   module y = IMR2A y
13   module ξy = IMR2 y.ξ
14   module f = IMR2A f
15   in record
16     { v = f.v
17     ; eqf = begin f.v ∘ f.ξx.f ∘ u = ( sym-assoc )
18       ( f.v ∘ f.ξx.f ) ∘ u = ( f.eqf ) ∘ (refl )
19       f.ξy.f ∘ u
20     ; eqφ = begin [ id , f.v ] ∘ [ u , id ] ∘ f.ξx.φ = ( sym-assoc ∘ ([ [-,-] ]-commute ) ∘ (refl) ∘ assoc )
21       [ u , id ] ∘ [ id , f.v ] ∘ f.ξx.φ = ( (refl) ∘ ( f.eqφ ) ∘ sym-assoc )
22       ([ u , id ] ∘ f.ξy.φ ) ∘ f.v
23     }
24 ; identity = λ {A} → refl
25 ; homomorphism = λ {X} {Y} {Z} {f} {g} → refl
26 ; F-resp = λ x → x
27 }
28
```

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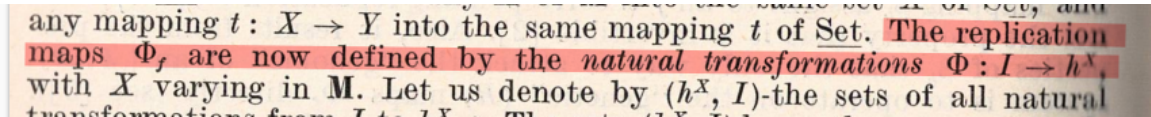
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any mapping  $t : X \rightarrow Y$  into the same mapping  $t$  of Set. The replication maps  $\Phi_f$  are now defined by the natural transformations  $\Phi : I \rightarrow h^X$ , with  $X$  varying in  $\mathbf{M}$ . Let us denote by  $(h^X, I)$ -the sets of all natural transformations from  $I$  to  $h^X$ .

# Coherent MR systems

$\Phi$  is pointed by  $f$ ; naturality means

commutes whenever  $\begin{array}{ccc} X & \xrightarrow{h} & Y \\ \ell \downarrow & & \downarrow r \\ X' & \xrightarrow{k} & Y' \end{array}$  is an arrow  $h \rightarrow k$  in the category  $\mathcal{C}^{[0 \rightarrow 1]}$ .

$$\begin{array}{ccc} Y & \xrightarrow{\Phi_h} & [A, Y] \\ r \downarrow & & \downarrow [A, r] \\ Y' & \xrightarrow{\Phi_k} & [A, Y'] \end{array}$$



# Towards the Rosen fibration

The profunctor of MR systems | 

**MRS**( $A, B$ ) is a profunctor  $\mathcal{C}^{op} \times \mathcal{C} \rightarrow \mathbf{Set}$ ; its action on morphisms  $u : A' \rightarrow A$ ,  $v : B \rightarrow B'$  is

$$\begin{aligned} \mathbf{MRS}(A, B) &\longrightarrow \mathbf{MRS}(A', B') \\ (A \rightarrow B, \Phi) &\longmapsto (A' \rightarrow A \rightarrow B \rightarrow B', [u, \text{cod}] \circ \Phi) \end{aligned}$$

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```
1 MRS-Profunctor : Bifunctor (Category.op C) C (Setoids (o u l u e) (o u l u e))
2 MRS-Profunctor = record
3   { F0 = λ { (A , B) → MR2-Setoid A B }
4   ; F1 = λ { {(A , B)} {(A' , B')} } (u , v) → record
5     { _{$}_ = λ {⟦ f , φ ⟧ → ⟦ v ∘ f ∘ u , (nHom u ∘r Cod) ∘v φ ⟧ }
6     ; cong = λ { ⟦ f , φ ⟧ } { ⟦ g , φ' ⟧ } (f≈g , φ≈φ') →
7       (∘-resp≈ Equiv.refl (∘-resp≈ f≈g Equiv.refl))
8       , (λ {x} → ∘-resp≈r (φ≈φ' {x}))
9     }
10 }
```

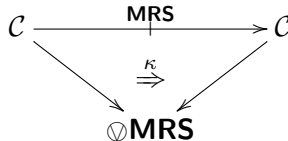
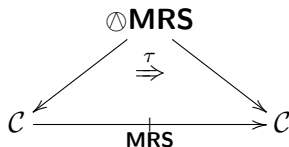
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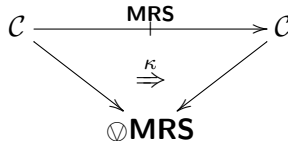
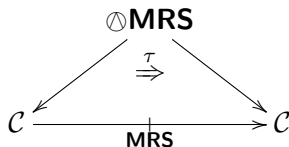
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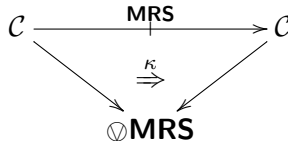
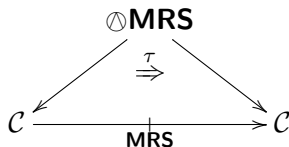


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- The category of **elements**  $\mathcal{E}(\mathbf{MRS})$  via pullback;

The two-sided discrete fibration  $\mathcal{C} \leftarrow \bigoplus \mathbf{MRS} \rightarrow \mathcal{C}$  is the Rosen fibration of the title.

# Rosen fibration(s)

Arbib<sup>1</sup> showed each MR system determines a **Mealy automaton** (input  $A$ , output  $B$ , state  $[A, B]$ ), functorially:

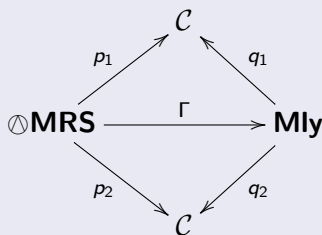
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There is a total category of Mealy automata and a fibred functor



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# Rosen fibration(s)

## Fibration of repair maps

The presheaf  $Rep : \mathcal{C}^{op} \rightarrow \mathbf{Set}$ ,  $A \mapsto \mathbf{Cat}(\mathcal{C}^{[0 \rightarrow 1]}, \mathcal{C})(cod, [A, cod])$  has the **fibration of repairs**  $\mathcal{E}(Rep)$  as its category of elements.

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**Objects:**  $(A, \Phi : cod \Rightarrow [A, cod])$ ; **morphisms**  $u : (A, \Phi) \rightarrow (A', \Upsilon)$  making  $\Phi = \Upsilon * [u, cod]$ .

$$\begin{array}{ccc} & cod & \\ \Upsilon \swarrow & & \searrow \Phi \\ [A', cod] & \xrightarrow{[u, cod]} & [A, cod] \end{array}$$

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$$\begin{array}{ccc} & cod & \\ \Upsilon \swarrow & & \searrow \Phi \\ [A', cod] & \xrightarrow{[u, cod]} & [A, cod] \end{array}$$

The fibration of repairs is a **discrete, coreflective subfibration** of the tabulator:

$$\mathcal{E}(Rep) \begin{array}{c} \xleftarrow{k} \\ \xrightarrow{\mathbb{T}} \\ \xleftarrow{\mathbb{J}} \end{array} \mathcal{T}(\mathbf{MRS}) \quad k(A, B; \xi @ (f, \Phi)) = (A, \Phi)$$

Rosen and others<sup>2 3 4</sup>

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$$A \longrightarrow B \longrightarrow [A, B] \longrightarrow [B, [A, B]] \longrightarrow \dots$$

defined by a **Fibonacci rule**  $M_0 = A$ ,

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defined by a **Fibonacci rule**  $M_0 = A$ ,  $M_1 = B$ ,  $M_n = [M_{n-2}, M_{n-1}]$ .

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<sup>2</sup>M. W. Warner, "Representations of (M,R)-systems by categories of automata". Bull. Math. Biology 44, 661–668 (1982).

<sup>3</sup>R. Rosen, "Anticipatory Systems..." Pergamon Press (1985).

<sup>4</sup>A. H. Louie, "More than life itself..." Ontos Verlag (2009).



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This 'vindicates Rosen': it makes *precise*, in *modern terms*, an idea he had decades ago.

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# Thanks!